Applied

(1). Accuracy compares the the degree of exactness or closeness of a mesurement to the expected or desired result. Whereas Precision is the consistency of an instrument output for a given value input.

(2) Data
- Expected value — 80V — Y_n
- Measured value — 79V. — X_n
- percentage accuracy — ?

- $q = 100\% - \%$ error

or

$q = A \times 100\%$ where A - Rel Accuracy

$$A = 1 - \left| \frac{Y_n - X_n}{Y_n} \right| = 1 - \left| \frac{80 - 79}{80} \right|$$

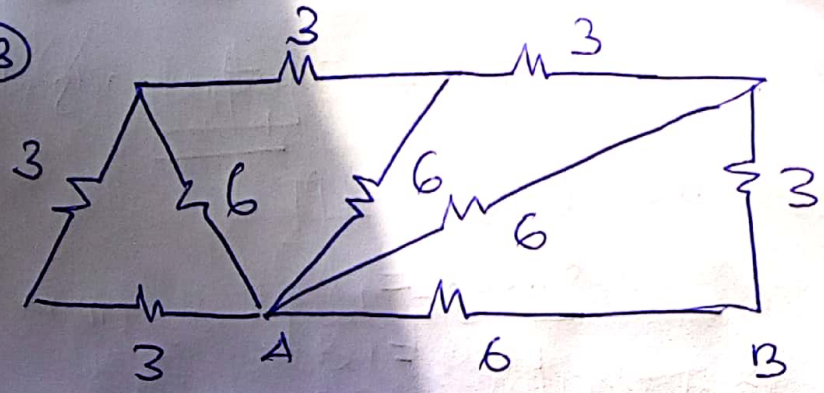
$$= \frac{79}{80}$$

(a) $= \frac{79}{80} \times 100\% = 98.75\%$ or

0.9875

Scanned with CamScanner

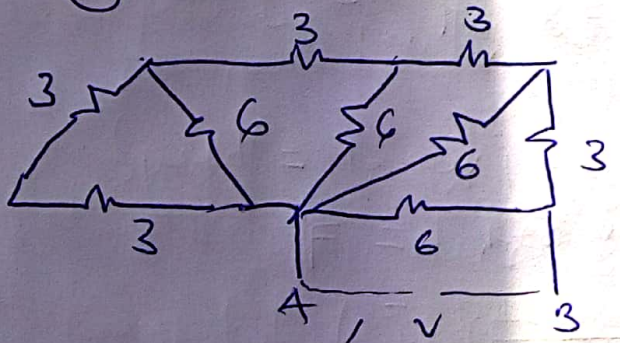
③



Soln

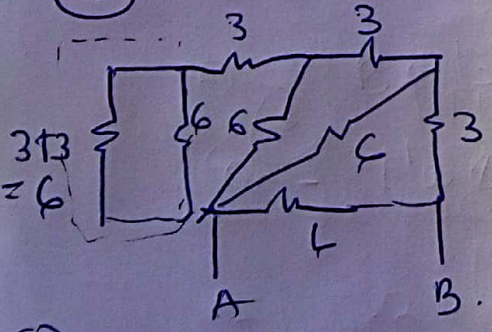
— See a voltage across the terminals AB and Calculate the resistance including the resistor between terminal A.

①



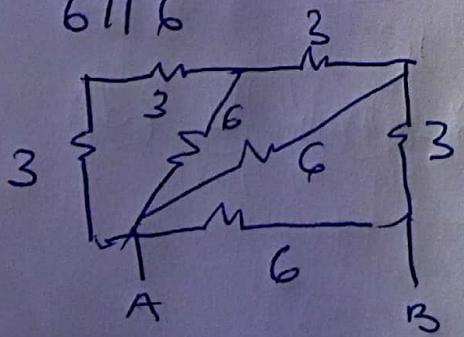
Imaginary.

②



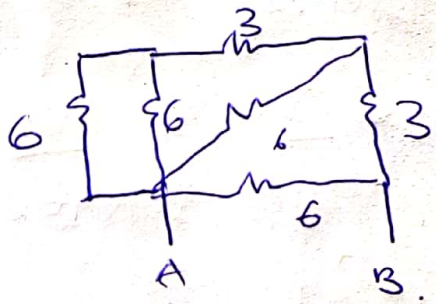
③ from the left

6||6



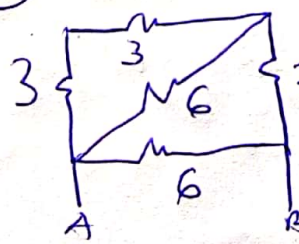
From the left of figure ③

3 + 3 — series.



From the left of fig ③

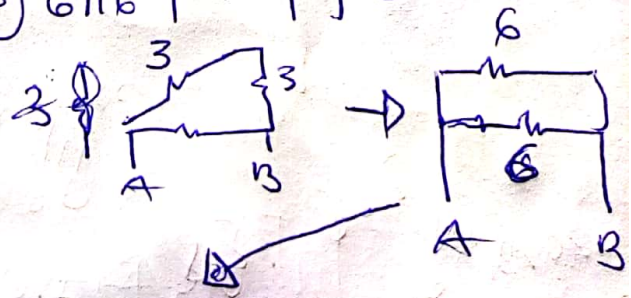
④ 6||6



⑤



⑥ 6||6 from fig ⑤



6||6 = 3 Ω

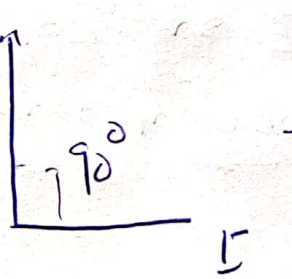
Scanned with CamScanner

④ A pure inductor does not consume any real power. Explain.

Sol.

Real power (P) = $VI\cos\varphi$

NB Current and voltage are 90° degrees out of phase.



– pure inductor.

Hence $P = VI\cos\varphi$ where $\varphi = 90°$

$P = VI\cos\varphi = VI\cos 90° = 0$

OR

In a pure inductive circuit current is lagging by 90° from voltage.

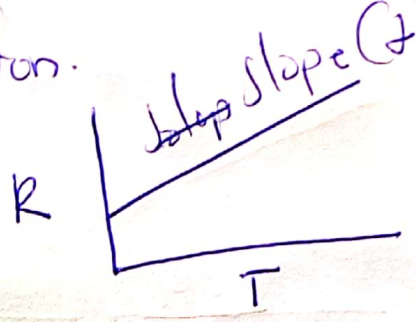
---

other Notes.

- Real power → The power actually consumed due to resistive Load (kW)
- Apparent power – power the grid must be able to withstand (VA)

Scanned with CamScanner

(5) Explanation.



$$R = R_0[1 + \alpha(\bar{T} - \bar{T}_0)] \quad \cdots (1)$$

R – Resistance at temp $\bar{T}$

R_0 – Reference Resistance at temp $\bar{T}_0$

$\bar{T}$ – Temperature in degrees

$\bar{T}_0$ – Reference temperature (usually 20° or 0°)

α – Temp Coefficient of resistance.

– Data,

$R = 48.5\ \Omega$ | $\bar{T} = ?$, $\bar{T}_0 = 20°$ | $\alpha = 0.004$

$R = 45\ \Omega$

– Plug in the data value into eqn (1)

$$48.5 = 45[1 + 0.004(\bar{T} - 20)]$$

– Now solve for $\bar{T}$

$$\frac{48.5}{45} = [1 + 0.004(\bar{T} - 20)]$$

$$1.078 = 1 + 0.004\bar{T} - 0.08$$

$$1.078 = 0.92 + 0.004\bar{T}$$

$$\bar{T} = \frac{0.158}{0.004} = 39.5°$$

Hence

Scanned with CamScanner

(6) Data

flux (ϕ) $= 5 \times 10^{-2}$ wb.

Area(A) $= 0.2\ m^2$.

Reluctance = ?

$$S = \frac{L}{\mu A}$$

where $\mu = \mu_r \mu_0$

$\mu_r = 1$.

$\mu_0 = 4\pi \times 10^{-7}$.

$L = 2\pi R_m$.

Area(A) $= 0.2 m^2$.

~~Area =~~

Considering the poles to be circular.

Area $= \pi r^2$.

$0.2 = \pi r^2$

$r_m = \sqrt{\frac{0.2}{\pi}}$

hence

$L = 2\pi \times \sqrt{\frac{0.2}{\pi}}$

e

Nos

$$S = \frac{2\pi \times \sqrt{\frac{0.2}{\pi}}}{0.2 \times 4\pi \times 10^{-7}} =$$

$= 6367831.305$

Please cross check sirs.

Scanned with CamScanner

⑦ Why is the star point of the supply transformer earthed/grounded -

Sol

Ideal there maybe several reason why.

-But

① It is grounded to conduct leakage current to the earth in the event of fault.

-other - state if you know -

(8) Data

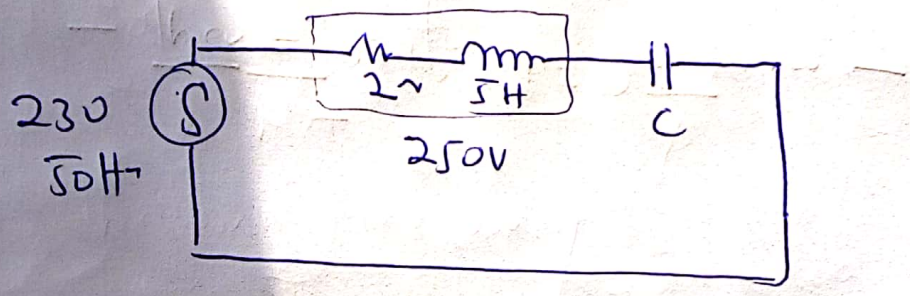
$V = 230V \quad f = 50Hz.$

$R = 2\Omega$

$L = 5H$

V_c – Voltage across coil – 250v.

– Diagram visualized.



How to go about it.

(1) Find Impedance of coil

(2) find current in coil = Total current

why – because of the series connection b/n the coil and the capacitor.

(3) Use total current to find Z from the circuit.

(4) Equate the Z_T in (3) to the Z_T seen from the circuit

(5) Now Solve for C.

Scanned with CamScanner

① $Z_c = \sqrt{R^2 + X_L^2} = \sqrt{2^2 + (2\pi \times 50 \times 5)^2}$

$= 1570.79\Omega$

$V_c = 250$

Hence $I_c = |I_{coil}| = \frac{V_c}{Z_c} = \frac{250}{1570.79} = 0.159A.$

But $I_c = I_T$ (total current) $= 0.159A.$

Stem ③

$$Z_{Total} = \frac{V_{Total}}{I_{Total}} = \frac{230}{0.159} = 1446.54\Omega.$$

④ $Z_{Total} = \sqrt{R^2 + (X_L - X_c)^2}$

$= \sqrt{2^2 + (1570.79 - X_c)^2}$

but $Z_{total} = 1446.54$

$1446.54 = \sqrt{2^2 + (1570.79 - X_c)^2}$

Square both sides.

$2.092 \times 10^6 = 4 + (1570.79 - X_c)^2.$

Expand

$X_c^2 - 3141.58 X_c + 378004 = 0.$

Solve for X_c $X_c = 3016.24$ or 125.34

Use $X_c = 125.34$ because X_c cannot be > Z_{total}.

Scanned with CamScanner

Now

$$X_C = \frac{1}{2\pi 50 \times C}$$

Why we are find capacitance C not ~~Impedance~~ Reactance X_C.

$$125.34 = \frac{1}{2\pi \times 50 \times C}$$

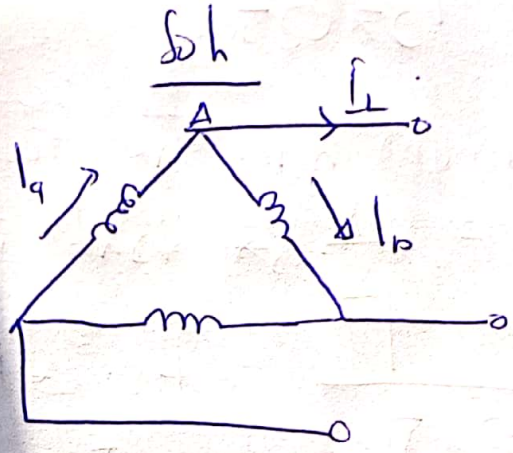
$$C = \frac{1}{2\pi \times 50 \times C}$$

$$= 25.394 F_{111}$$



Scanned with CamScanner

(9) Relationship between current and Relationship b/n voltages in a delta connected system.



(1) Current.

- Line current I_L is $\sqrt{3}$ times the phase current

Mathematically

$$I_L = \sqrt{3} \times I_p.$$

Wkt, take KCL at node A.

$$I_L = I_a - I_b.$$

$$= I\angle 0^\circ - I\angle -120^\circ$$

$$= I(1 + j0) - I(-0.5 - j0.866)$$

$$= I\left(\tfrac{3}{2} + j\tfrac{\sqrt{3}}{2}\right) = \sqrt{3}\, I\angle 30^\circ.$$

(2) Voltage.

- Line voltage is equal to phase voltage

$$V_L = V_p.$$

Scanned with CamScanner

(Nn) From 10 - 15

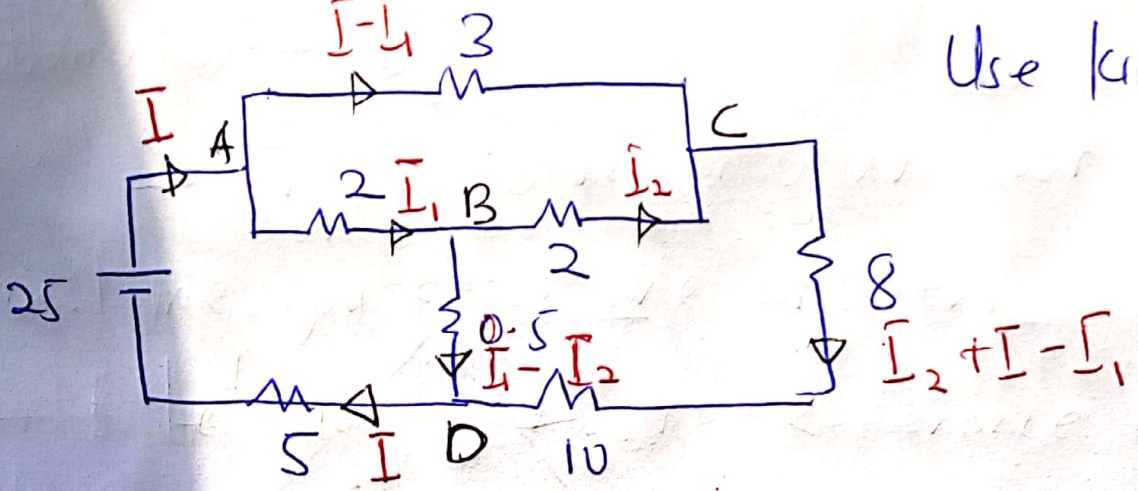
— Try to Solve on your own

Trust me you know the answers to that

— Let Google and the Reference books be your guide

— lets See the part B

(1)



Use Kirchhoff

① Assign currents correctly on the branches. ✓

② Take loops

$\mathcal{E}_{emf} + \mathcal{E}_{IR} = 0$

loop ABCA

(NB) Take a loop in each window

$\mathcal{E}_{emf} = \mathcal{E}_{IR}$

loop ABCA

$0 - 2I_1 - 2I_2 + 3(I - I_1) = 0$

$3I - 5I_1 - 2I_2 = 0$ - ①

loop ABDA

$25 - 2I_1 - 0.5(I_1 - I_2) - 5(I) = 0$

$-5I - 2.5(I_1) + 0.5I_2 = -25$ - ②

loop BCDB

$0 - 2I_2 - 8(I_2 + I - I_1) - 10(I_2 + I - I_1) + 0.5(I_1 - I_2) = 0$

$-18I + 18.5I_1 - 29.5I_2 = 0$ - ③

- Solve ① ② and ③ simultaneously

$I = 3.63, I_1 = 2.55, I_2 = 0.93$

to loop ABCA.

$0 = 2I_1 + 2I_2 - 3(I - I_1)$

$0 = -3I + 5I_1 + 2I_2$ - ①

loop ABDA

$25 = 2I_1 + 0.5(I_1 - I_2) + 5(I)$

$25 = 5I + 2.5I_1 - 0.5I_2$ - ②

loop BCDB

$0 = 2I_2 + 8(I_2 + I - I_1) + 10(I_2 + I - I_1) - 0.5(I_1 - I_2)$

$0 = 18I - 18.5I_1 + 29.5I_2$ - ③

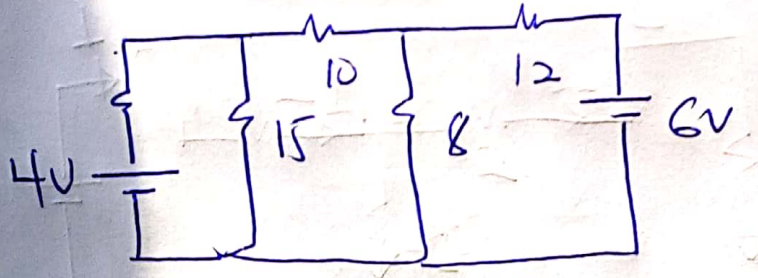
- Solve ① ② and ③

$I = 3.63, I_1 = 2.55, I_2 = 0.93$

Scanned with CamScanner

(2) — Check in the mid sem solution.

(3)



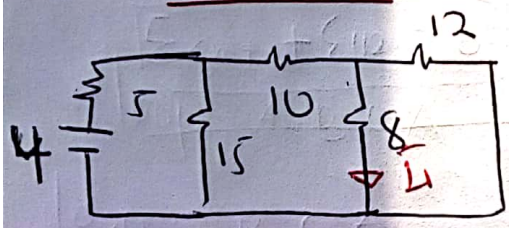
— Use superposition to find current in 8Ω

— ~~Use~~ let 4V act alone — find current I_8

Let 6V act alone — find current I_8

Sum the two current to find ~~I_8~~ I_8.

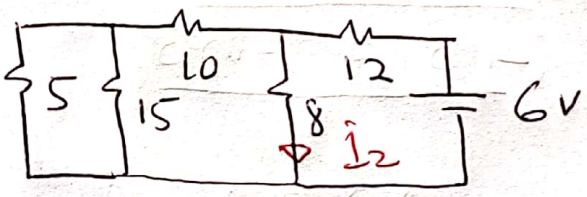
4V alone



Now use any method [Kirchhoff, ohm's law, mesh, nodal ~~and~~ & or Find total resistance ~~and use~~ use that to find total current and apply current division to find I_1

$$I_1 = 0.097A$$

6V alone



— Same here

$$I_2 = 0.222A$$

(NB) The current I_1 and I_2 are in same direction

Hence

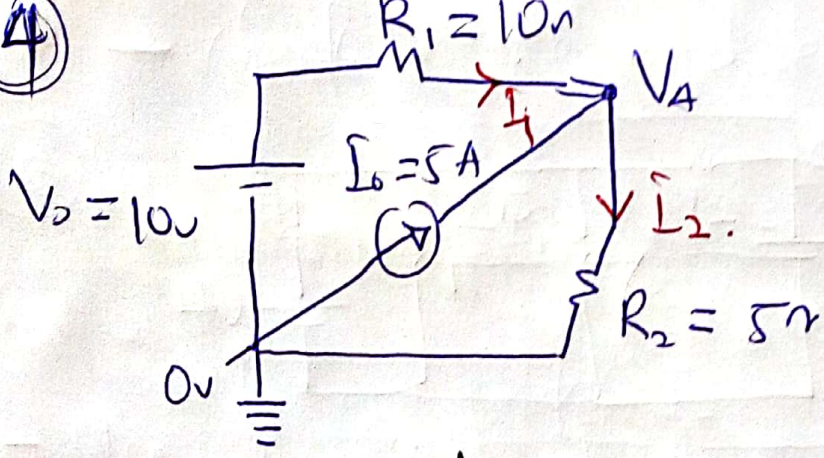
$$I_8 = I_1 + I_2$$

$$= 0.097 + 0.222$$

$$= 0.319A$$

Scanned with CamScanner

(4)



— Use nodal analysis to find I_1, I_2 and V_A

NB Nodal = KCL + ohms law.

$$I = \frac{V}{R}.$$

(1) We have two node. 0v and V_A.

Make KCL on V_A.

$$I_1 + 5A = I_2.$$

Write the $\frac{V}{R}$ of the above.

$$I_1 = \frac{10 - V_A}{10}, \quad I_2 = \frac{V_A}{5}.$$

$$I_1 + 5A = I_2$$

$$\frac{10 - V_A}{10} + 5A = \frac{V_A}{5}.$$

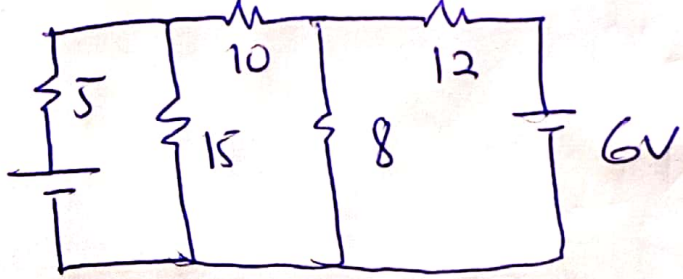
Solve for V_A = **20V** — V_A

Now $I_1 = \frac{10 - 20}{10}$ $^{-V_A}$ = **-1A.** — I_1

$$I_2 = \frac{20}{5} : \frac{V_A}{5} = \frac{20}{5} = \textbf{4A} \quad - I_2$$

Scanned with CamScanner

(5)

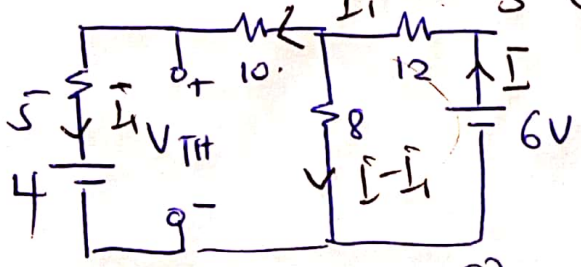


— Use Thevenin to find I_{15}.

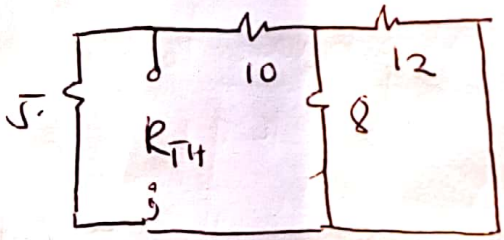
Ans → $I_{15} = 0.1192A$

$V_{TH} = 3.6$

$R_{TH} = 3.737\Omega$







$V_{TH} - 4 = 5I_1$ or — (2)

$V_{TH} = -10I_1 + 8(I - I_1)$ — (1)

What do you need to find I and I_1, plug the back in (1) or (2) to find V_{TH}

In finding I and I_1 take loops.

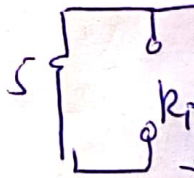
$4 = -5I_1 - 10I_1 + 8(I - I_1)$

$4 = 8I - 23I_1$ — (3)

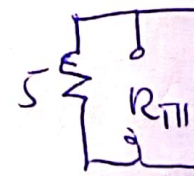
$6 = 12I + 8(I - I_1)$

$6 = 20I - 8I_1$ — (4)

Solve for I and I_1, put in equation 1 or 2 to find V_{TH} ~~3.35V~~ = 3.6 →



$10 + (8 \| 12)$



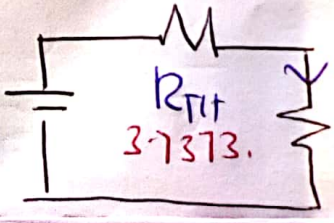
$4 \quad 8 \quad 10$

$R_{TH} = 5 \| 14.8$

$= \frac{5 \times 14.8}{5 + 14.8}$

= Get value.

= ~~2.4444~~ 3.737



Scanned with CamScanner

(5)

$$I = \frac{53}{198}$$

$$I_1 = -\frac{8}{99}$$

$$V_{TH} = 5\left(-\frac{8}{99}\right) + 4$$

$$= 3.5959$$

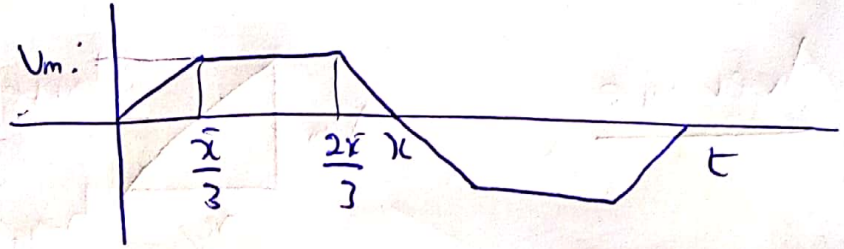
$$= 3.6$$

$$R_{TH} = 3.7373$$

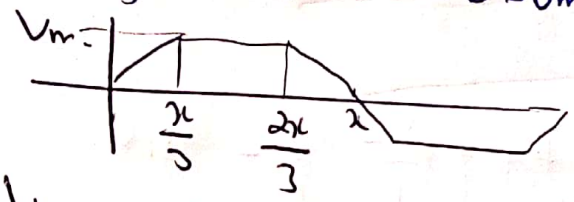
$$I_{15} = \frac{3.6}{3.73 + 15} = 0.192\ A$$

Scanned with CamScanner

(6)



Average voltage 0.666 Vm



$V_{av} = \frac{\text{Area under 1 cycle}}{\text{Period } (T)}$

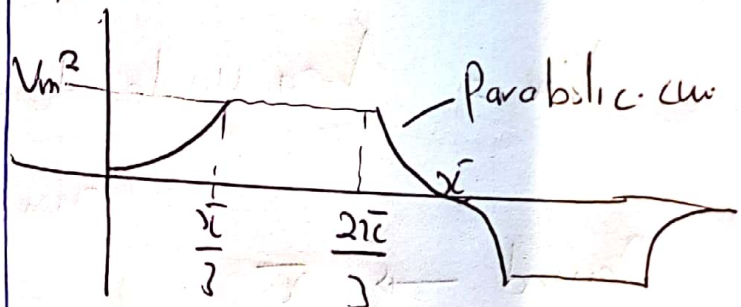
or

$V_{av} = \frac{1}{T}\int_0^T F(t)\,dt.$

What do we do?

RMS should be 0.745 Vm.

- Square the above wave first



- Parabola and rectangle.

$V_{rms} = \sqrt{\frac{1}{T}\int_0^T [F(t)]^2\,dt}$

or

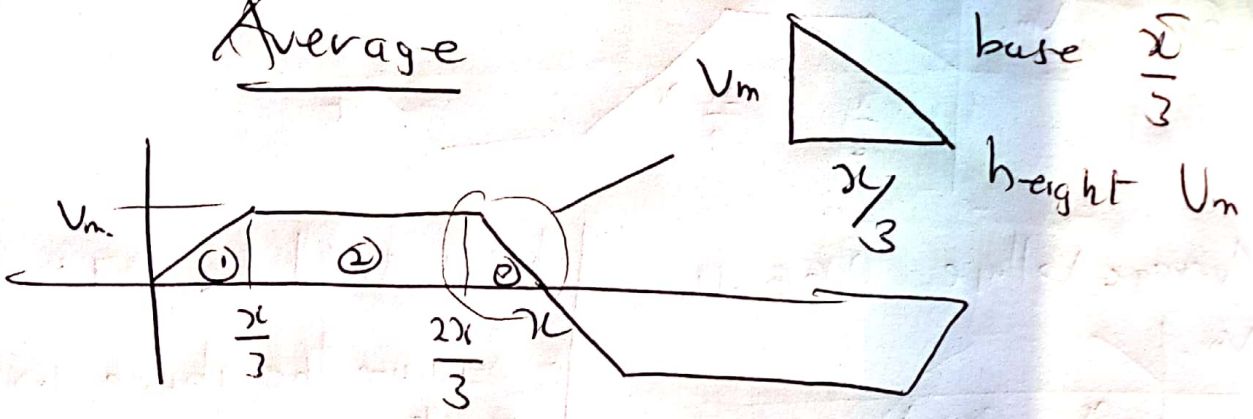
$V_{rms} = \sqrt{\frac{\text{Area Under Squared Curve}}{\text{Period } (T)}}$

What do we do?

Next Page.

Scanned with CamScanner

Average



$$V_{av} = \frac{\text{Area under 1 complete cycle}}{\text{Period}}$$

Period $\rightarrow \pi$

$$= \frac{\text{Area ①} + \text{Area ②} + \text{Area ③}}{\pi}$$

$$= \frac{\left(\frac{1}{2} \times V_m \times \frac{\pi}{3}\right) + \left(V_m \times \frac{\pi}{3}\right) + \frac{1}{2} \times V_m \times \frac{\pi}{3}}{\pi}$$

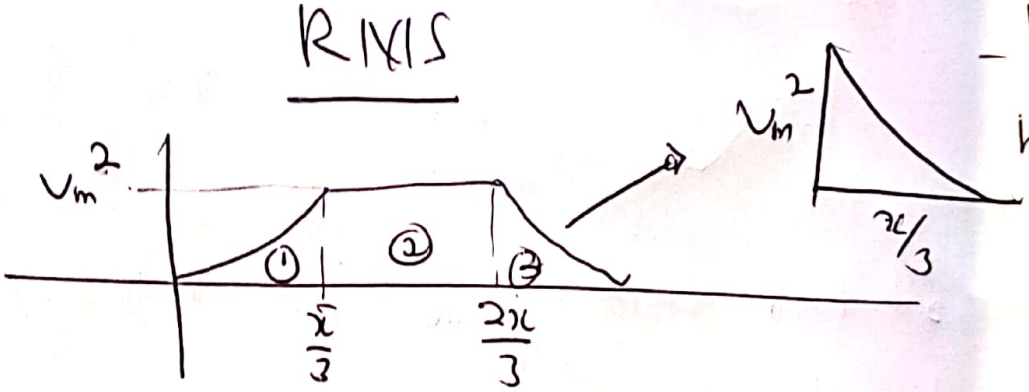
$$= \frac{\frac{1}{2}\left[\pi + \frac{\pi}{3}\right] \times V_m}{\pi}$$

$$= \frac{\frac{2}{3}\pi}{\pi} \times V_m$$

$$= 0.666\, V_m \text{ ///}$$

Scanned with CamScanner

RMS



- base $\pi/3$
height V_m^2

- Area of parabola - $\frac{1}{3} \times b \times h$.

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T f(t)^2 dt} = \sqrt{\frac{\text{Area under Squared curve}}{T}}$$

$T = 3\pi$

$$V_{rms} = \sqrt{\frac{\text{Area ① + Area ② + Area ③}}{3\pi}}$$

$$V_{rms} = \sqrt{\frac{\left(\frac{1}{3} \times \frac{\pi}{3} \times V_m^2\right) + \left(V_m^2 \times \frac{\pi}{3}\right) + \left(\frac{1}{3} \times \frac{\pi}{3} \times V_m^2\right)}{3\pi}}$$

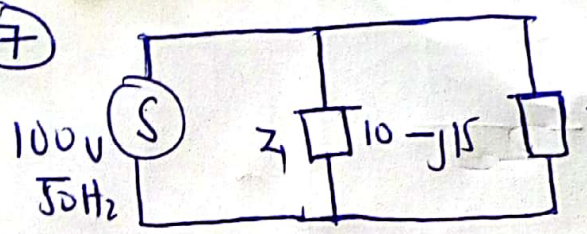
$$V_{rms} = \sqrt{\frac{\frac{2}{9}\pi V_m^2 + \frac{\pi}{3} V_n^2}{3\pi}}$$

$$= \sqrt{\frac{5}{9}} \times \sqrt{V_n^2} = \sqrt{\frac{5}{9}} \times V_m.$$

$$V_{rms} = 0.745\, V_m$$

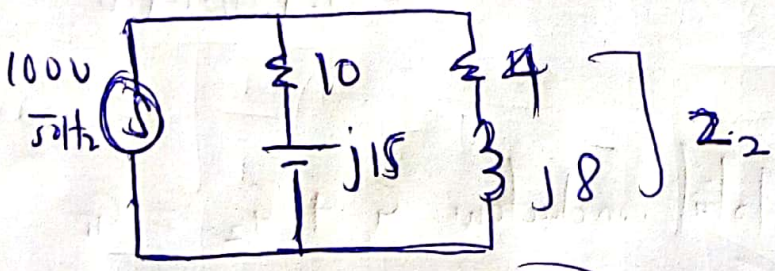
Scanned with CamScanner

(7)



Calculate Admittance, Conductance, Susceptance, (b/a) current and power factor

The circuit can be redrawn as.



Why

Impedance $Z = R + jL$

$Z = R - jC$

Now

— find Impedance for Z_1 and Z_2

$$Z_1 = \sqrt{R^2 + X_C^2}$$

$$= \sqrt{10^2 + 15^2}$$

$$= 18.03\Omega$$

$$Z_1 = \sqrt{R^2 + X_L^2}$$

$$= \sqrt{4^2 + 8^2} = 4\sqrt{5}$$

$$= 8.94\Omega$$

Now find Conductance for Z_1 and Z_2.

Z_1

Conductance (G_1)

$$G_1 = \frac{R_1}{(Z_1)^2} = \frac{10}{(18.03)^2}$$

$$= 0.0308\text{℧}$$

Z_2

Conductance (G_2)

$$G_2 = \frac{R_2}{(Z_2)^2} = \frac{4}{(8.94)^2}$$

$$= 0.05004\text{℧}$$

Scanned with CamScanner

Now Also find Susceptance B

for Z_1 Z_2

$$B_1 = \frac{X_{C_1}}{(Z_1)^2} = \frac{15}{(18.03)^2}$$

$$= 0.0461 \text{ N}$$

$$B_2 = \frac{X_{L_2}}{(Z_2)^2} = \frac{8}{(8.94)^2}$$

$$= 0.1000 \text{ N}$$

Now

(i) Conductance = Total conductance $G_1 + G_2 = G_T$

$$= 0.0308 + 0.05004 = 0.08084$$

(ii) Total susceptance

$$= B_T = B_1 \pm B_2 = 0.0461 + 0.1000 \text{ N}$$

$$= 0.1461 \text{ N}$$

(iii) Total Admittance (Y_T)

$$= \sqrt{(G_T)^2 + (B_T)^2} = \sqrt{}$$

$$= \sqrt{(0.08084)^2 + (0.1461)^2}$$

$$= 0.167 \text{ N}$$

(iv) Total Current

$$I_T = V Y_T$$

$$= 100 \times 0.167$$

$$= 16.698 \text{ A}.$$

(v) Power factor

$$\cos\phi = \frac{|G_T|}{|Y_T|} = \frac{0.08084}{0.167}$$

$$\cos\phi = \frac{0.08084}{0.167}$$

Scanned with CamScanner

(8) Data

$N = 200$

$A = 250 m^2$

$B = 0.5 T$

Speed = 1200 r.p.m.

(a) Equation of instantaneous induced emf:

$\therefore e = E_m \sin \omega t$.

E_m = Maximum value = $N\omega\phi$

$\omega = 2\pi F$.

$\theta = BA$; $B = \theta / A$.

Speed = 1200 r.p.m.:

F = Rev per Second

$= \frac{1200}{60} = 20 rps$.

$\omega = 2\pi \times 20 = 125.67 rad/sec.$

$\theta = BA = 0.5 \times 250 = 125 wb.$

$E_m = N\omega\phi = 200 \times 125.67 \times 125.$

$= 3141750$

Hence

$e = 3141750 \sin(125.67t)$

(b) Maximum value of induced emf.

$E_m = N\omega\theta$

$= 3141750 V$

Scanned with CamScanner

(i) $\theta = \omega t = 90^0$

$e = 314150 \sin(90) = 314150\ V$

(ii) $\theta = \omega t = 0^0$

$e = 314150 \sin 0 = 0\ V$

(iii) $\theta = \omega t = 45^0$

$e = 314150 \sin 45$

= evaluate ____ V

(10) Part 4

line voltage V_L = 400V.

phase voltage $V_\phi = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 231$ Volts

Load impedance $Z_\phi = (8 + j6)$

$= 10 \angle 36.9^\circ \Omega$.

phase lag angle $\phi = 36.9^\circ$ lag.

(1) Line current $(I_L) = I_\phi = \frac{V_\phi}{Z_\phi} = \frac{231}{10}$

$= 23.1$ A

(2) power factor $= \cos\phi = \cos 36.9^\circ = 0.8$ lag.

or. $\cos\phi = \frac{|R_\phi|}{|Z_\phi|} = \frac{8}{10} = 0.8$.

(3) Total power $= \sqrt{3} \times V_L = \sqrt{3} \times 400$

$= \sqrt{3} \times V_L \times I_L \cos\phi$

$= \sqrt{3} \times 400 \times 23.1 \times \cos 36.9$

(4) Power factor if it was delta connected.

— Try — (00)

Q.9 and 11

Try on your own

(12) A 6KVA. =

Data!

- Motor input = 6KVA. = P
- Original power factor (p.f) = $\cos\phi_1 = 0.7$
- final power factor (p.f) = $\cos\phi_2 = 0.89$

find ϕ and ϕ_2

$\phi_1 = \cos^{-1}(0.7) = 45.57$

$\phi_2 = \cos^{-1}(0.89) = 27.13$

But not that

$KVA \times p.f = KW$

$p.f = \frac{KW}{KVA}$

$\tan(\phi_1) = \tan(45.57) = 1.020$

$\tan(\phi_2) = \tan(27.13) = 0.5124$

Required KVAr of a capacitor to improve for 0.7 → 0.89

$P \frac{KVA}{KVAR} = \phi(\tan\phi_1 - \tan\phi_2)$

$= 6KVA(1.020 - 0.5$

Scanned with CamScanner

$P = \vec{KVAR} = 3.054$ KVAR.

Now Rating of ~~e~~ capacitor.

$$\frac{P}{3}$$

$$\frac{3.054}{3} = 1.018 \text{KVAR}$$

NB

~~A~~ I am not sure of queston (12)'s answer - So try doing some more reading — I have to so now.

Thank You

Don't focus on this alone it's just pasco — But know how to Solve all.

~~I saw another paper on~~

All The best